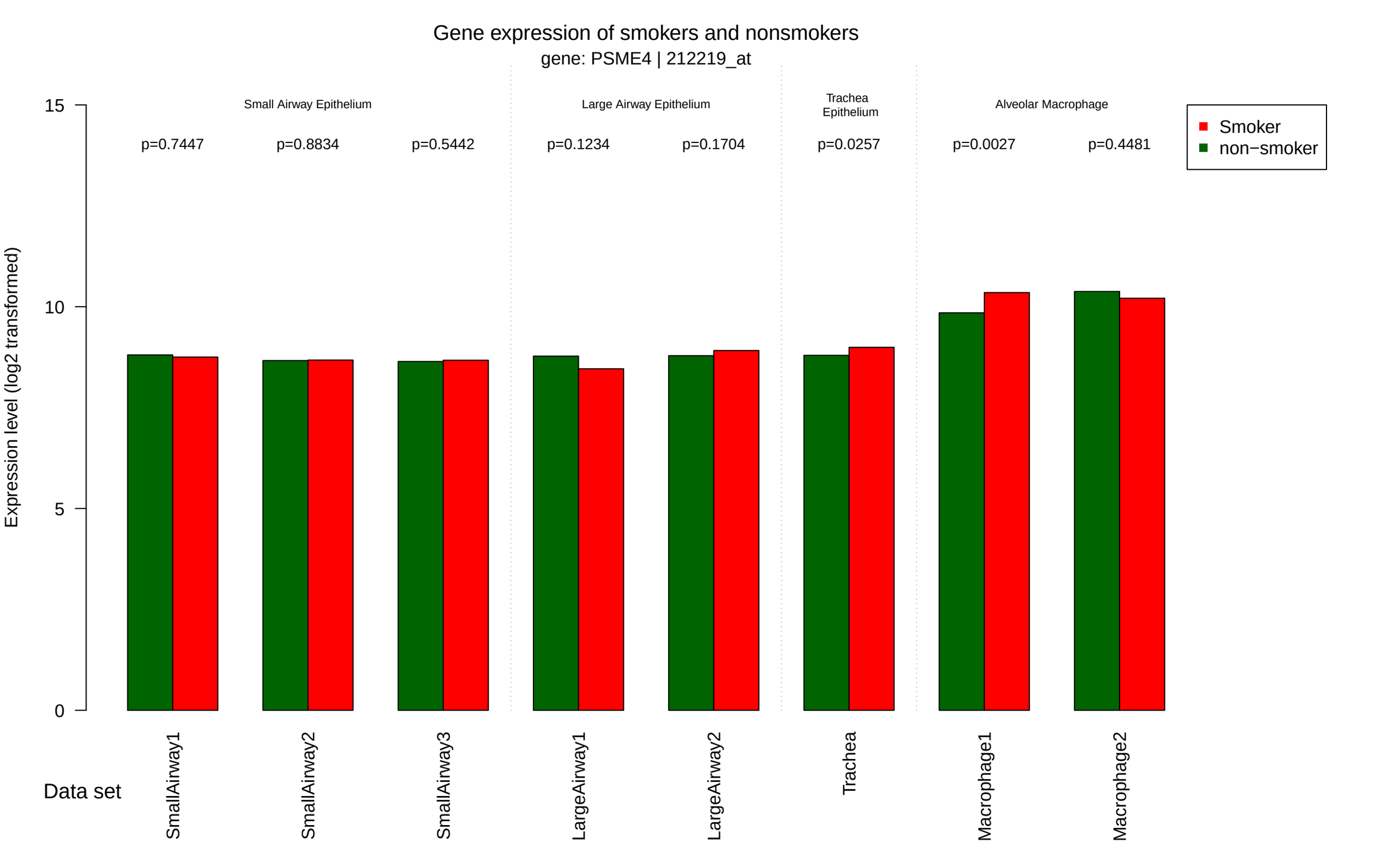
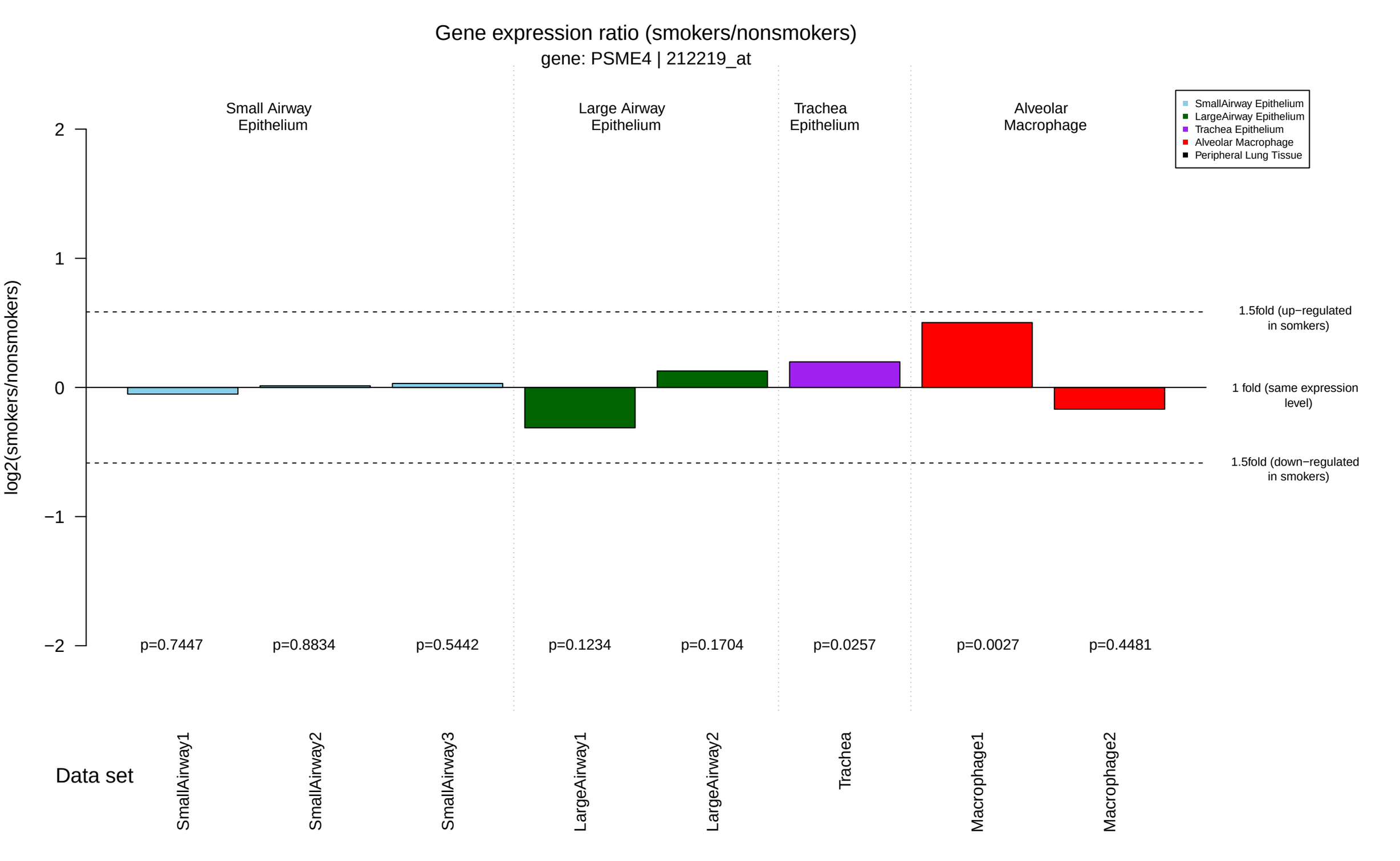
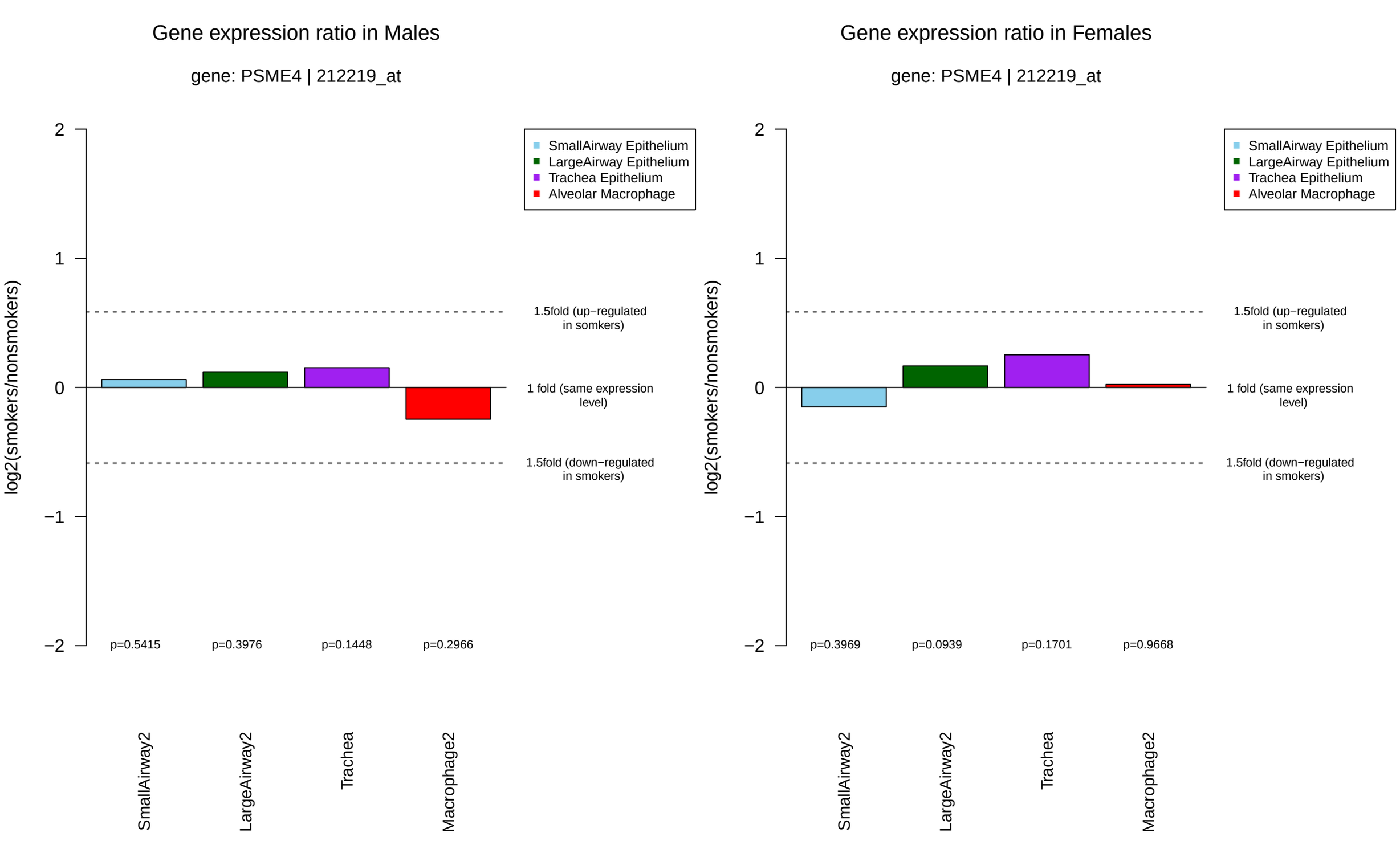
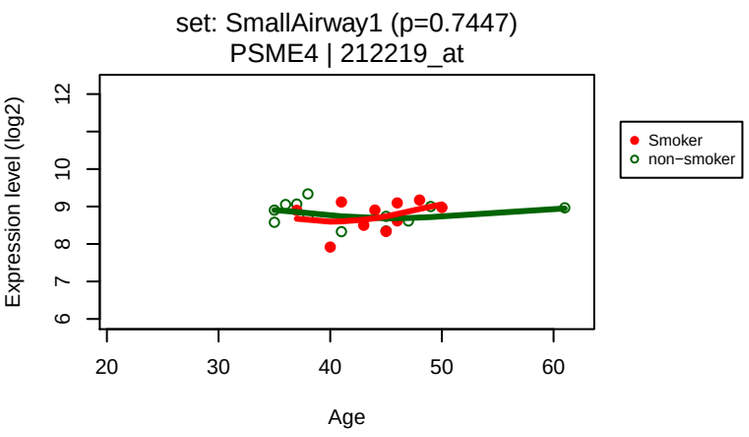
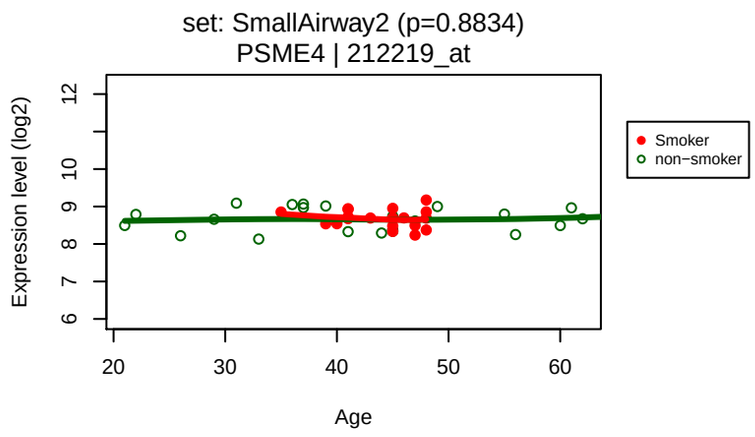
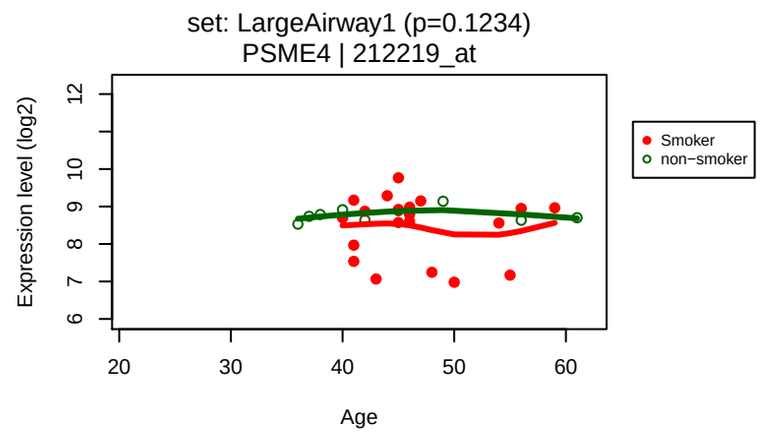

Gene expression of smokers and nonsmokers
gene: PSME4 | 212219_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7447
p=0.8834
p=0.5442
p=0.1234
p=0.1704
p=0.0257
p=0.0027
p=0.4481
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

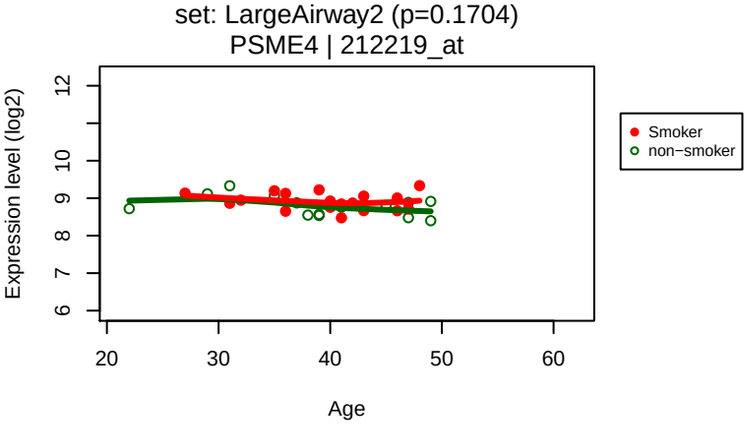

Gene expression ratio (smokers/nonsmokers)
gene: PSME4 | 212219_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7447
p=0.8834
p=0.5442
p=0.1234
p=0.1704
p=0.0257
p=0.0027
p=0.4481
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

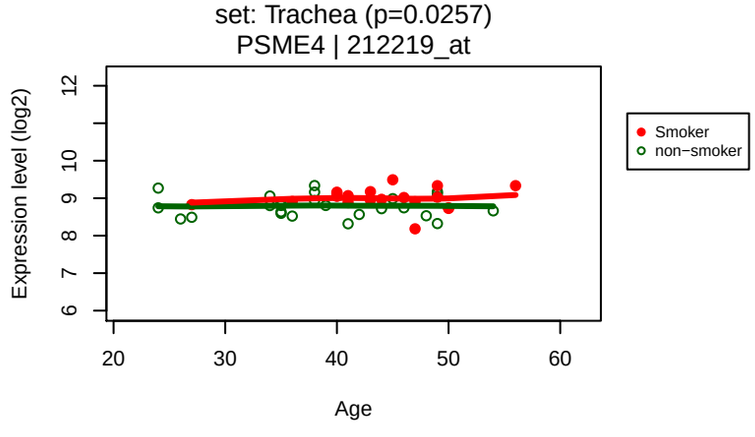

Gene expression ratio in Males
gene: PSME4 | 212219_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5415
p=0.3976
p=0.1448
p=0.2966
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PSME4 | 212219_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3969
p=0.0939
p=0.1701
p=0.9668
SmallAirway2
LargeAirway2
Trachea
Macrophage2

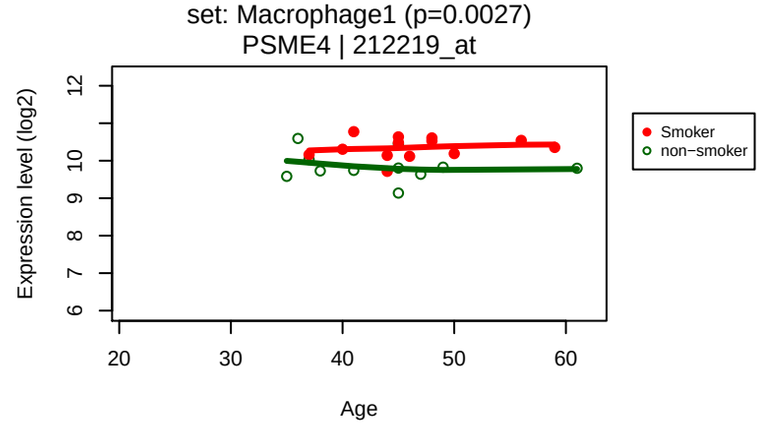

set: SmallAirway1 (p=0.7447)
PSME4 | 212219_at
Expression level (log2)
Age
Smoker
non−smoker

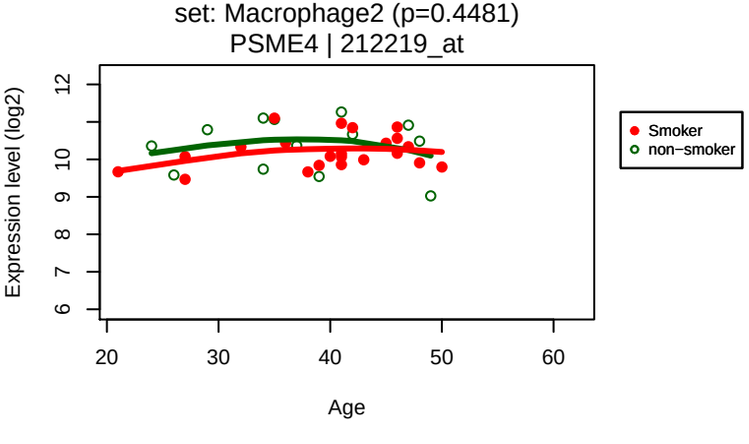

set: SmallAirway2 (p=0.8834)
PSME4 | 212219_at
Expression level (log2)
Age
Smoker
non−smoker

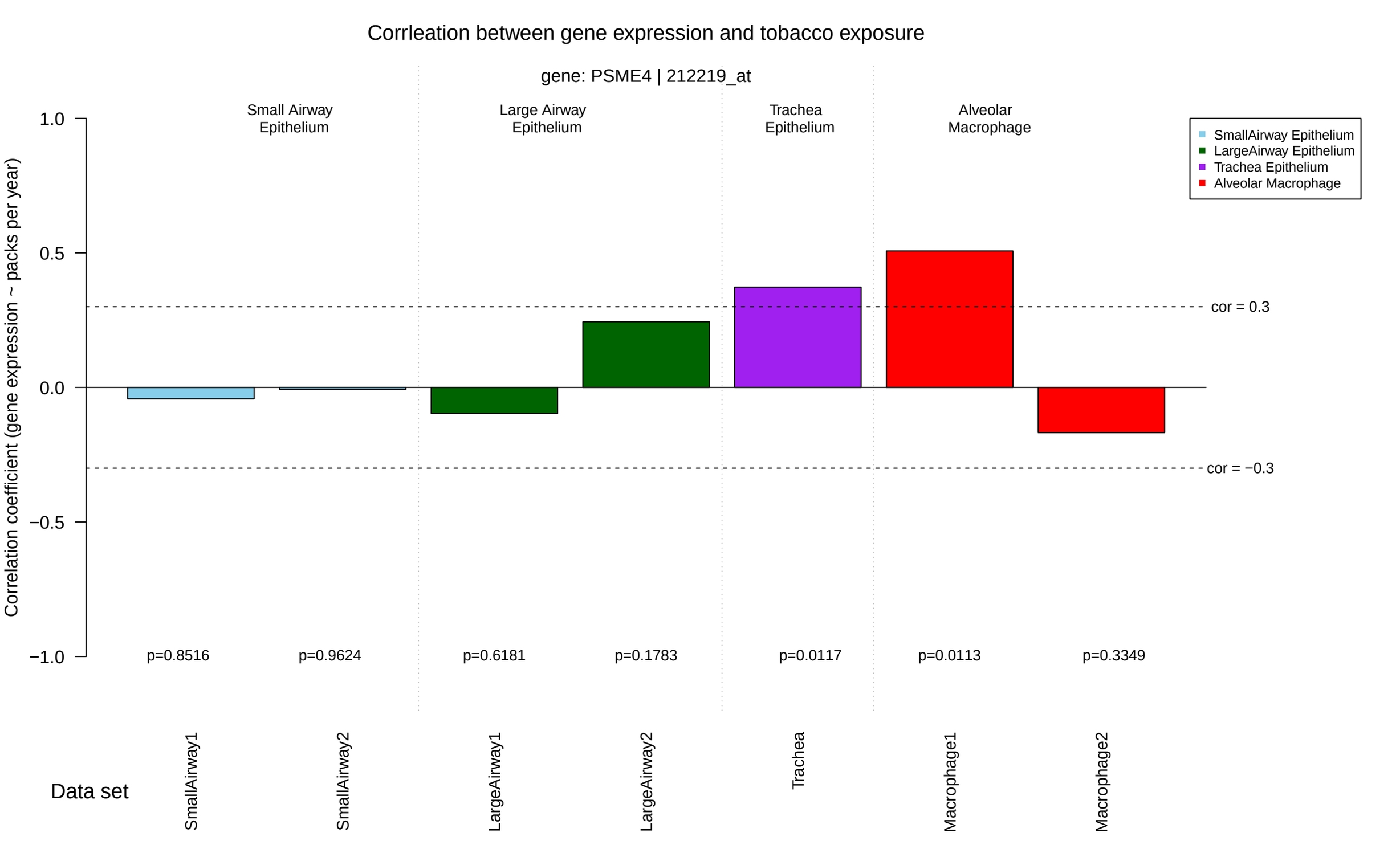

set: LargeAirway1 (p=0.1234)
PSME4 | 212219_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1704)
PSME4 | 212219_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0257)
PSME4 | 212219_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0027)
PSME4 | 212219_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4481)
PSME4 | 212219_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PSME4 | 212219_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8516
p=0.9624
p=0.6181
p=0.1783
p=0.0117
p=0.0113
p=0.3349
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2